FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – II

PAPER-2

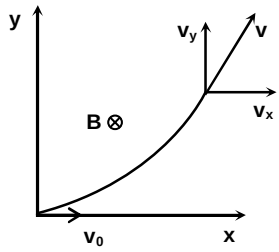
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	C	24.	D	47.	A
2.	C	25.	B	48.	C
3.	D	26.	C	49.	D
4.	B, C	27.	D	50.	A, B, C, D
5.	A, B, C, D	28.	B, C	51.	B, C
6.	A, C, D	29.	B	52.	C, D
7.	A, B, C, D	30.	C, D	53.	A, B, C
8.	B, D	31.	A, B, C, D	54.	A, C
9.	B	32.	B	55.	B
10.	A	33.	B	56.	C
11.	2	34.	4	57.	4
12.	4	35.	4	58.	0
13.	1	36.	5	59.	1
14.	4	37.	4	60.	2
15.	7	38.	3	61.	2
16.	2	39.	6	62.	2
17.	2	40.	3	63.	0
18.	8	41.	9	64.	4
19.	7	42.	5	65.	9
20.	2	43.	2	66.	2
21.	00369.31	44.	00003.33	67.	00009.50
22.	00002.50	45.	00063.75	68.	00000.80
23.	00006.25	46.	00009.25	69.	00000.50

Physics

PART – I

SECTION – A

1. $\vec{F} = q(v_x\hat{i} + v_y\hat{j}) \times B_0\left(1+\frac{y}{d}\right)(-\hat{k}) = F_x\hat{i} + F_y\hat{j}$



$|F_x| = qB_0\left(1+\frac{y}{d}\right)v_y$

$-\int_{v_0}^{v_x} m dv_x = q\int_0^d B_0\left(1+\frac{y}{d}\right)dy \quad \left(\frac{dv_x}{dy} \text{ is negative}\right)$

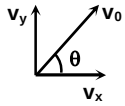
$v_x = v_0 - \frac{qB_0}{m}\left(\frac{3}{2}d\right)$

$v_x = \frac{3qB_0d}{m} - \frac{3}{2}\frac{qB_0d}{m} = \frac{3}{2}\frac{qB_0d}{m} = \frac{v_0}{2}$

$v_x^2 + v_y^2 = v_0^2$

$\frac{v_0^2}{4} + v_y^2 = v_0^2$

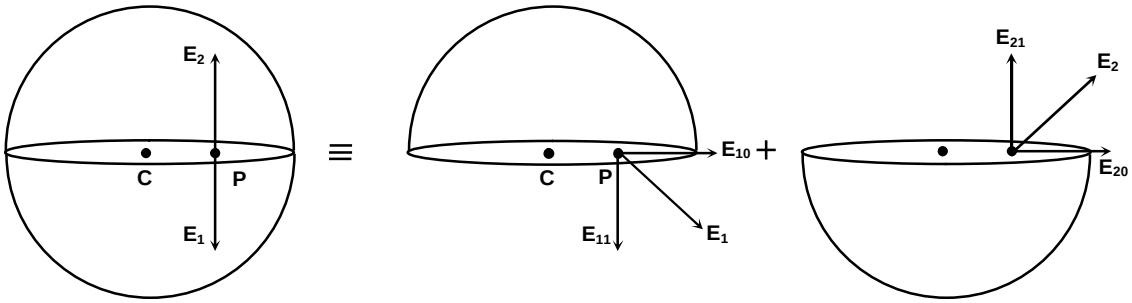
$\Rightarrow \quad v_y = \frac{\sqrt{3}v_0}{2}$



$\tan\theta = \frac{v_y}{v_x} = \sqrt{3}$

$\theta = 60°$

2. E_1 and E_2 are the magnitude of electric field due to upper hemispherical shell and lower spherical shell respectively, whose components along the diameter and its perpendicular direction are shown in the figure. By the combination of two hemispherical shells: At P, $E_{11} = E_{21}$ and $E_{10} = E_{20} = 0$
net electric field must be zero.

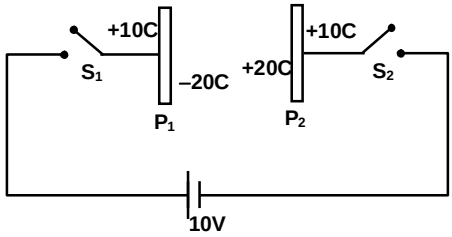


3. $\phi_{BCFG} = \phi_{CDGH} = \phi_{EFGH} = \frac{q}{24\varepsilon_0}$, $\phi_{EHAD} = 0$, $\phi_{ABEF} = \phi_{ABCD} = x$

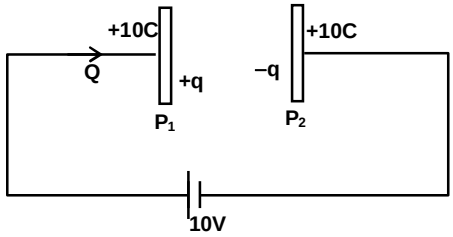
$2 \times x + 0 + \frac{q}{24\varepsilon_0} \times 3 = 0$

$\Rightarrow x = -\frac{q}{16\varepsilon_0}$

4.



When switches are not closed



When switches are closed

$$\frac{q}{A\varepsilon_0 \times d} = 10$$

$$\Rightarrow \quad q = 10 \times \frac{A\varepsilon_0}{d} = 20 \text{ C}$$

Q = 20 + 20 = 40 C

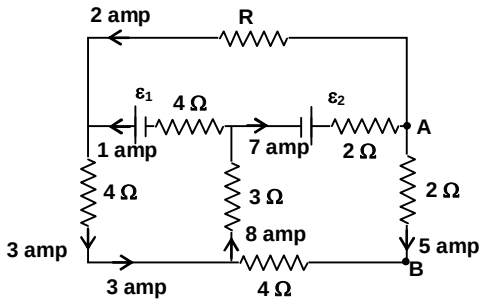
W_B = (20 + 20) 10 = 400 J

5. By applying Kirchhoff's law

ε_1 = 40 V

ε_2 = 68 V

R = 9 Ω and current through 4Ω = 5 amp.



6. As the ammeters are ideal, so

$$R_{eq} = \frac{R_1R_2R_3}{R_1R_2 + R_2R_3 + R_3R_1}$$

The readings of two ammeters are different so resistors R_1 and R_3 are different.

If R_1 and R_3 are interchanged, the reading of ammeters will also change.

If $R_2 = R_1$ (If R_2 and R_1 switched the reading of the ammeter do not vary) similarly if $R_2 = R_3$ (R_2 and R_3 switched the reading of ammeters do not change)

If $R_2 = R_1$, $I_1 = I_2 = 0.15$, $I_3 = 0.05$ and I = 0.35

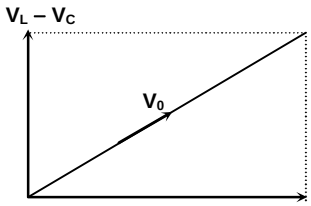
If $R_2 = R_3$, $I_2 = I_3 = 0.10$, $I_1 = 0.20$ and I = 0.40 amp

7. $V_R^2 + V_L^2 = V_1^2 = (20\sqrt{2})^2 = 800$ …(i)

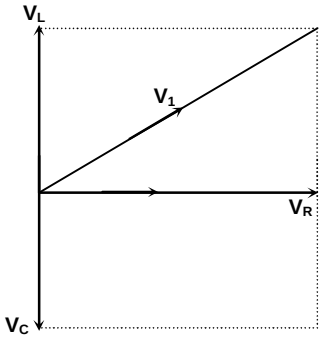
$V_L - V_C = 15$ …(ii)

$\sqrt{V_R^2 + (V_L - V_C)^2} = (25)^2$

$V_R^2 + 225 = 625$



V_0 = source voltage



$\Rightarrow V_R = 20$ volts

$V_L = 20$ volts

$V_L - V_C = 15$

$20 - V_C = 15$

$V_C = 5$ volts

Power factor = $\cos\phi = \dfrac{R}{Z} = \dfrac{iR}{iZ} = \dfrac{V_R}{V_Z} = \dfrac{20}{25} = \dfrac{4}{5}$

8. By applying Kirchhoff's Law

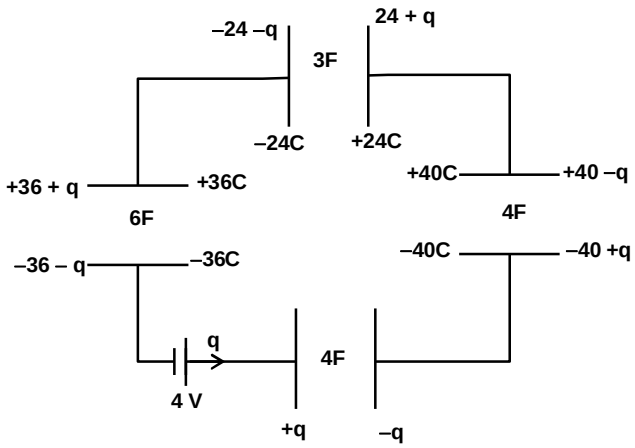
$$-4 + \frac{q}{4} + \frac{-40+q}{4} + \frac{24+q}{3} + \frac{36+q}{6} = 0$$

$$-4 + \frac{q}{4} - 10 + \frac{q}{4} + 8 + \frac{q}{3} + 6 + \frac{q}{6} = 0$$

$$\frac{q}{4} + \frac{q}{4} + \frac{q}{3} + \frac{q}{6} = 0$$

$\Rightarrow q = 0$

Hence, there is no flow of charge.



9-10. $dQ = dU + dW$

$\Rightarrow \quad nCdT = nC_v dT + PdV$

$$\Rightarrow \quad \frac{2RT}{T_0} - 3R = \frac{2RT}{V}\cdot\frac{dV}{dT}. \qquad \ldots(1)$$

From (1) $\displaystyle\int_{T_0}^{T}\left(\frac{1}{T_0} - \frac{3}{2T}\right)dT = \int_{V_0}^{V}\frac{dV}{V}$

On solving, $\displaystyle V = V_0\left(\frac{T_0}{T}\right)^{3/2} e^{\left(\frac{T-T_0}{T_0}\right)}$

$$\frac{2RT}{P} = V_0\left(\frac{T_0}{T}\right)^{3/2} e^{\left(\frac{T-T_0}{T_0}\right)}$$

$$\frac{1}{P} = \frac{V_0 T_0^{3/2}}{2RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$$

For minimum volume, $\dfrac{dv}{dT} = 0$

So, $\displaystyle T = \frac{3}{2}T_0$,

$$V_{min} = V_0\left(\frac{T_0}{\frac{3}{2}T_0}\right)^{3/2} e^{\left(\frac{\frac{3}{2}T_0 - T_0}{T_0}\right)}$$

$$V_{min} = \left(\frac{2}{3}\right)^{3/2} V_0 e^{1/2}$$

SECTION – C

11. $E = -\frac{\partial V}{\partial x} = \frac{3x^2}{6\varepsilon_0} = \frac{x^2}{2\varepsilon_0}$

$\phi = E.A. = \frac{x^2}{2\varepsilon_0} A$

$\frac{d\phi}{dx} = \frac{(xA)}{\varepsilon_0} \Rightarrow d\phi = \frac{(xAdx)}{\varepsilon_0}$

$\rho(x) = \frac{dq}{Adx} = \frac{(xAdx)}{(Adx)} = x$

$\rho(x = 2) = 2$

Alternating Method

$\nabla\vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \frac{x}{\varepsilon_0} = \frac{\rho}{\varepsilon_0} \Rightarrow \rho = x$

If $x = 2 \Rightarrow \rho = 2$

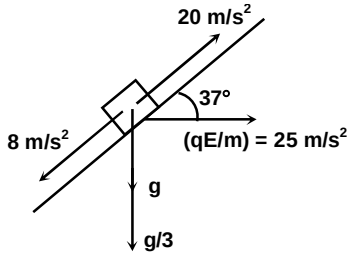
12. The rate of heat loss by cube of thickness Δx, side ℓ and made of density ρ is given by

$-ms\frac{dT}{dt} = \varepsilon\sigma A\left(T^4 - T_0^4\right)$

$-\frac{dT}{dt} = \frac{\varepsilon\sigma A(T^4 - T_0^4)}{ms}$

13. F.B.D. of block

$a_{net} = 12\ m/s^2$, up the incline plane



$S = ut + \frac{1}{2}gt^2$

$3.6 \times \frac{5}{3} = \frac{1}{2} \times 12 \times t^2$

$\Rightarrow t = 1\ sec$

14. Tension in upper rope $= \frac{800}{3} N.$

By balancing Torque about 'O'

$\frac{800}{3} = \frac{V_0^2\varepsilon_0 A}{2d^2} = \frac{V_0^2\varepsilon_0 20}{2\left(\frac{1}{5}\right)3\varepsilon_0}$

$V_0 = 4\,Volts$

15. $q = t^3 - t^2 + 4t$

$i = \frac{dq}{dt} = 3t^2 - 2t + 4$

$i(t = 1) = 5\,amp$

$\frac{di}{dt} = 6t - 2$

$\frac{di}{dt}(t = 1) = 4$

Current is increasing with time

$q(t = 1) = 1 - 1 + 4 = 4$ coulombs

$V_A - V_B = \frac{4}{2} + \frac{4}{2} + 5 \times \frac{3}{5} = 7$ Volts

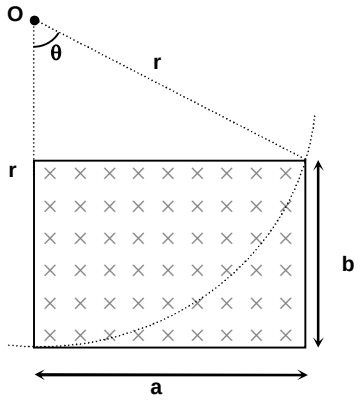
16. $r = \frac{mv}{qB}$

$\sin\theta = \frac{a}{r},\ \cos\theta = \frac{r-b}{r}$

$\left(\frac{a}{r}\right)^2 + \left(\frac{r-b}{r}\right)^2 = 1$

$\Rightarrow r = \frac{(a^2+b^2)}{2b} = \frac{mv}{qB}$

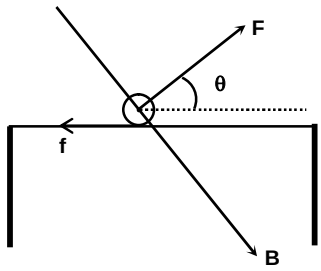
$v = \frac{qB(a^2+b^2)}{2mb} = 2\ \text{m/s}$



17. $F_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$

$i\ell B_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$

$B_{min} = 2$ Tesla



18. Induced emf = $\frac{B\omega\ell^2}{2}$, current $i_0 = \frac{B\omega\ell^2}{2R}$

Torque about P = $\tau = \int (idx)B.x = \frac{iB\ell^2}{2}$

$= \left(\frac{B\omega\ell^2}{2R}\right)\left(\frac{B\ell^2}{2}\right) = 8$

So, power P = $\vec{\tau}\cdot\vec{\omega}$ = 160 Watt

Alternate Solution

Power = Heat loss in resistance = $Ri_1^2 = \frac{B^2\omega^2\ell^4}{4R} = 160\ \text{Watt}$

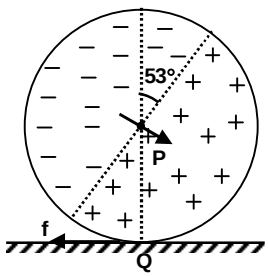
19. $\tau_Q = PE\sin 53°$

$= \left(\rho\frac{2}{3}\pi R^3\right)\left(\frac{3R}{4}\right)E_0\frac{4}{5}$

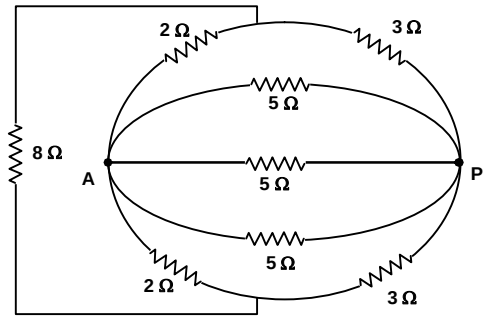
$= \frac{2}{5}\rho\pi R^4 E_0 = \left(\frac{7}{5}mR^2\alpha\right)$

$a_{cm} = R\alpha = \frac{2}{7}\frac{\rho\pi R^3 E_0}{m}$

$f = ma_{cm} = \frac{2}{7}\rho\pi R^3 E_0$



20. $R_{AB} = 1\ \Omega$ so $R_{AB} = 1 + 1 = 2\ \Omega$



SECTION – D

21. For first vessel.

$\Delta U = 0$

$\Delta Q_1 = \Delta W = nRT\ell n\left(\frac{V_f}{V_i}\right) = 1.R.300.\ell n2$

$= (0.6931) \times 300 \times R = 207.93\ R$

For second vessel $\Delta V = 0$

$\Delta Q_2 = \Delta U = nC_V\Delta T = 1 \cdot \frac{3}{2}R\Delta T$

$\Delta Q_1 = \Delta Q_2$

$207.93R = \frac{3}{2}R\Delta T$

$\Rightarrow \Delta T = 138.62$

$T_f = 300 + 138.62 = 438.62$

$U_{mix} = U_1 + U_2$

$(n_1 + n_2)\, C_{Vmix} T_{mix} = n_1 C_{V_1} T_1 + n_2 C V_2 T_2$

$T_{mix} = \frac{T_1 + T_2}{n_1 + n_2} = 369.31K.$

22 $X_C = \frac{V_C}{I} = \frac{1000}{1} = 1000\Omega$

Current lags behind voltage

So, box has an inductor

Power factor = 0.8 = $\frac{R}{Z} = \frac{800}{\sqrt{(800)^2 + (X_L - 1000)^2}}$

$X_L = 1600\ \Omega$

$\omega L = 1600$

$L = \frac{1600}{2 \times 3.2 \times 100} = 2.5\ H$

23. Impulse, $F\Delta t = m\sqrt{2gh}$

$(i\ell B)\Delta t = m\sqrt{2gh}$ …(i)

$i\Delta t = \Delta q$ …(ii)

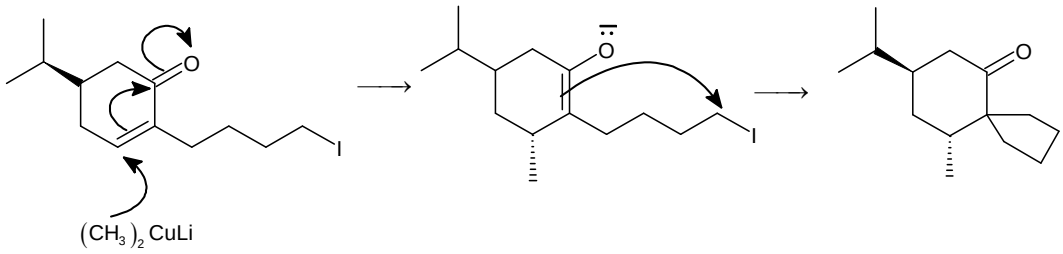
$\Delta q = \frac{m\sqrt{2gh}}{B\ell} = \frac{5\sqrt{2 \times 10 \times 5}}{2 \times 4} = 6.25$ Coulomb

Chemistry

PART – II

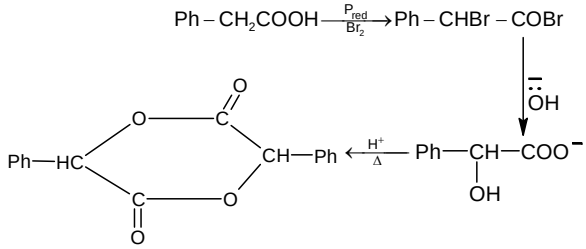
SECTION – A

24.

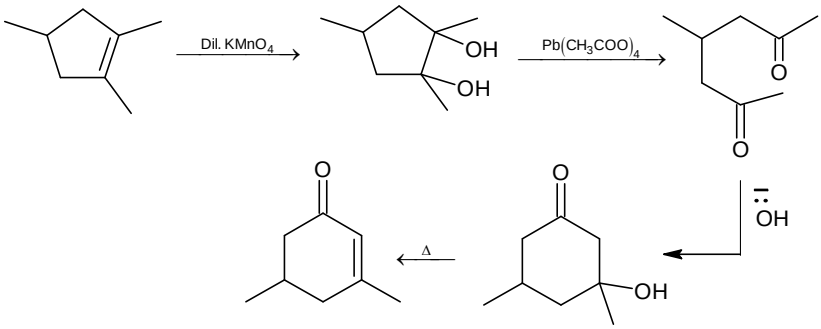


Lower side of plane

25.

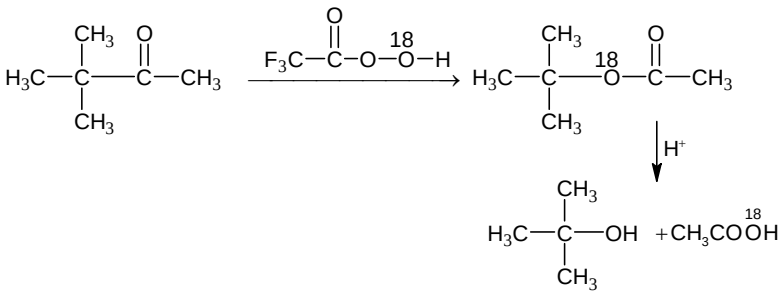


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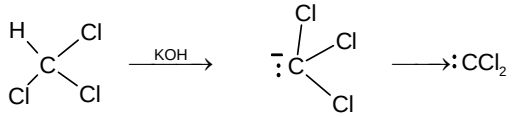


27. PhCOOH is more acidic than H_2CO_3 while PhOH is less acidic than H_2CO_3.

28.



29.



30.

$HCHO \xrightarrow{\text{Fehling solution}} HCOOH$

$PhCHO \xrightarrow{\text{Fehling solution}} \text{No reaction}$

$HCHO + NH_3 \longrightarrow (CH_2)_6 N_4$

$PhCHO + NH_3 \longrightarrow Ph-HC(OH)(NH_2) \xrightarrow{\Delta} PhCH=NH$

$HCHO + NaOH \longrightarrow CH_3OH + HCOO^-$

$PhCHO + NaOH \longrightarrow PhCH_2OH + PhCOO^-$

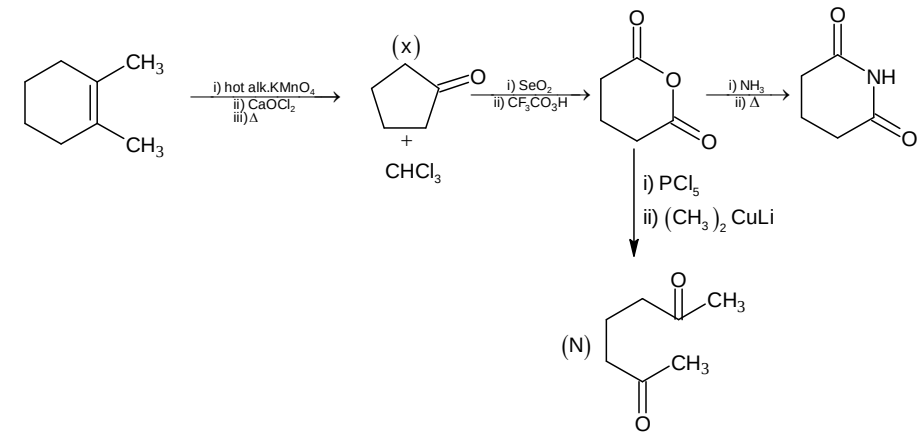
$HCHO \xrightarrow{\text{Tollens}} HCOO^-$

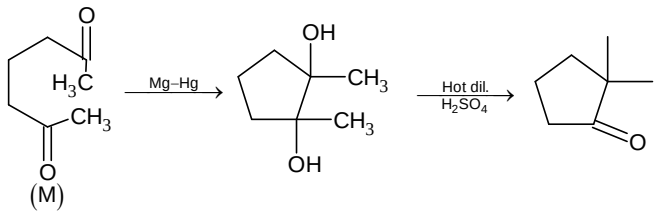
$PhCHO \xrightarrow{\text{Tollens}} PhCOO^-$

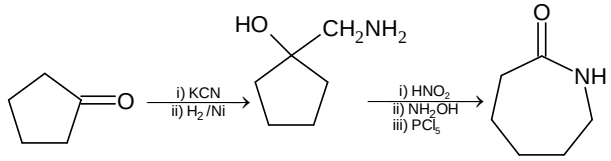
31.

$$CH_3-CH_2-C\equiv N \xrightarrow{H_3O^+} CH_3-CH_2-COOH \xrightarrow[\text{ii) } H_3O^+]{\text{i) } Cl_2/P_{red}} H_3C-\underset{Cl}{CH}-COOH \xrightarrow{NH_3} H_3C-\underset{NH_2}{CH}-COOH$$

33.



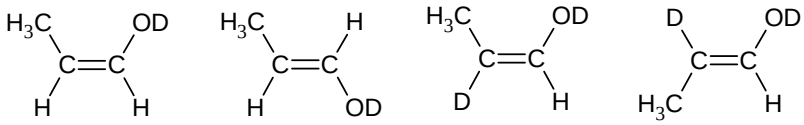




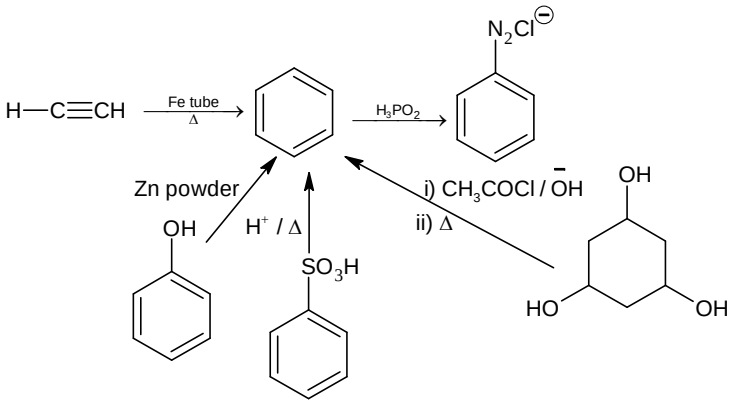
Caprolactum

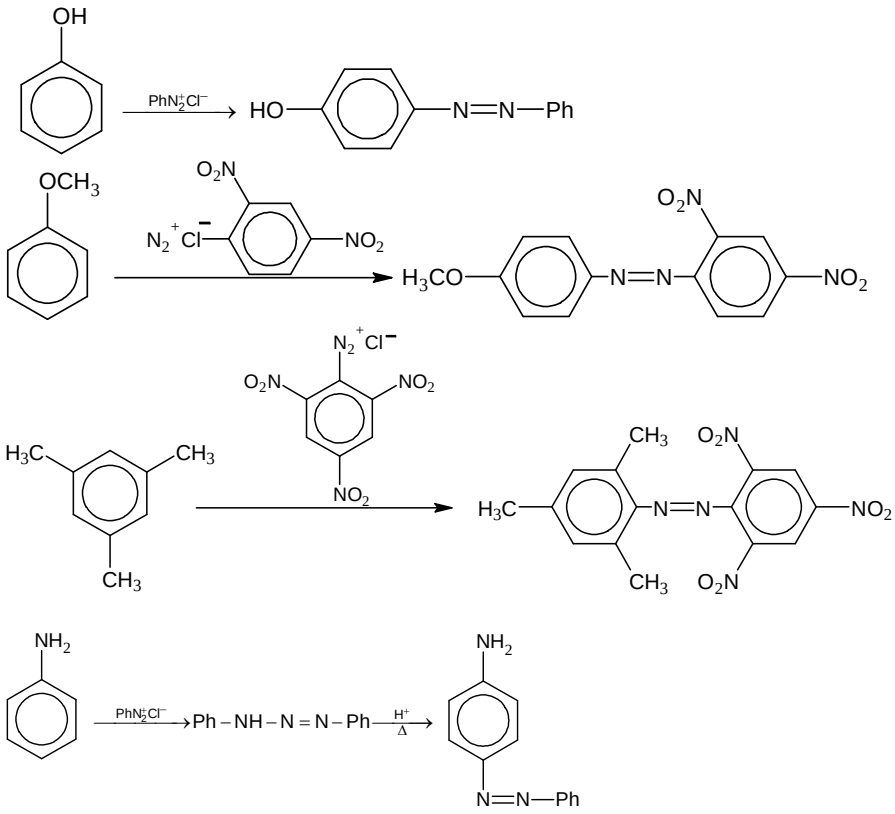
Polymer of caprolactum → Nylon – 6

SECTION – C

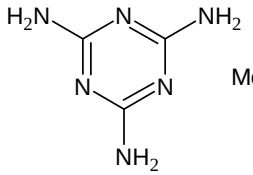
34. 

35. 

36. 

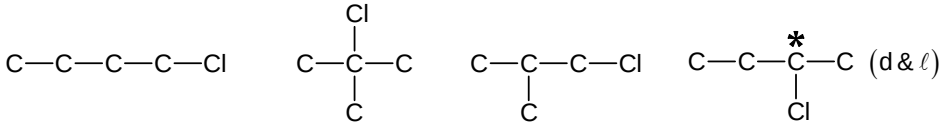
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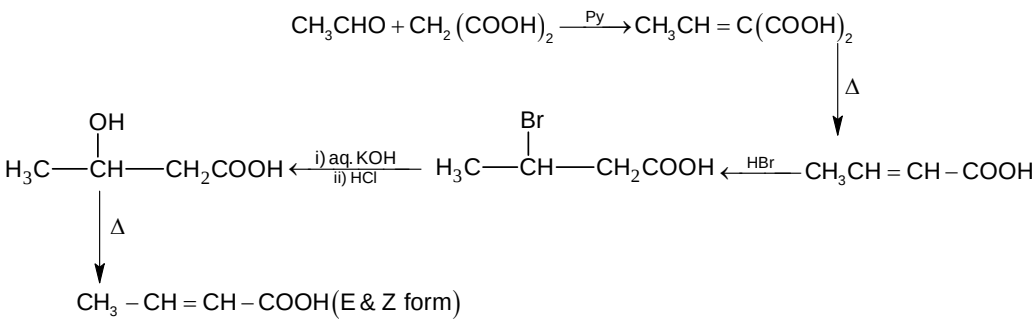
38. Glucose, Fructose and Mannose.

39.  Melamine

40. Polyethylene, Polystyrene, Poly(methyl(methacrylate)

41. $\frac{160}{56} \times 3.15 = 9$

42. 

43. 

SECTION – D

44. $\alpha - D \longrightarrow x\%$

$\beta - D \longrightarrow (1-x)$

$90^o = (110x - 30x) + 30^o$

$x = \frac{60}{80} = \frac{6}{8}$

$\underset{1-x}{\alpha - D} \rightleftharpoons \underset{x}{\beta - D}$

$K_{eq} = \frac{x}{1-x}$

$\therefore K'_{eq}$ for $\beta - D \rightleftharpoons \alpha - D$ is $\frac{1}{K_{eq}} = \frac{1-x}{x}$

$\therefore K'_{eq} = \frac{2/8}{6/8} = \frac{2}{6} = \frac{1}{3} = 3.33$

$\therefore 90^o = \frac{(x)(110) + (1-x)30^o}{1}$

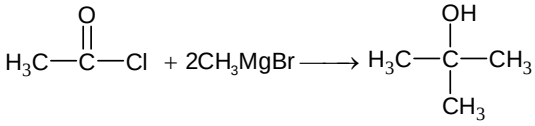
45. Mili moles of H_2SO_4 initially = 70 × 0.5 = 35
Mili moles of NaOH required = 65 × 0.5 = 32.5
$\therefore$ Mili moles of H_2SO_4 consumed = 35 – (32.5/2)
= 18.75
$\therefore$ Mili moles NH_3 = 37.5
$\therefore W_{NH_3} = \frac{37.5 \times 17}{1000} = 0.6375$ gm

$$\%NH_3 = \frac{0.6375}{1.0} \times 100 = 63.75$$

46. Moles of $CH_3COCl = \frac{78.5}{78.5} = 1.0$ mole

Moles of $CH_3MgBr = \frac{29.75}{119} = 0.25$

$\therefore$ CH_3MgBr is limiting reagent.



Moles of alcohol = $\frac{0.25}{2}$

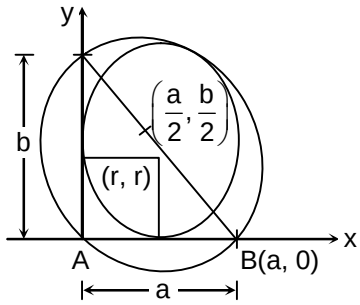
Weight of alcohol = $\frac{0.25}{2}$ × 74 = 9.25 gm

Mathematics

PART – III

SECTION – A

47. Equation of $S = \left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2 + b^2}{4}$



$x^2 + y^2 - ax - by = 0$
And circle with centre (r, r) and radius r
Touches AB and AC its equation is
$(x - r)^2 + (y - r)^2 = r^2$
$x^2 + y^2 - 2xr - 2yr + r^2 = 0$
Both circle touches internally or externally
Using $\sqrt{\left(r - \frac{a}{2}\right)^2 + \left(r - \frac{b}{2}\right)^2} = \sqrt{\left(\frac{a^2}{4} + \frac{b^2}{4}\right)} \pm r$

$\Delta ABC = \frac{r_1 r_2}{4}$

48. Equation of tangent at (i, y_i)
$y = (2i + b)x - i^2 + c$
Also at $(i + 1, y_{i+1})$, $y = (2(i + 1) + b)x - (i + 1)^2 + c$
Point of intersection is $x = \frac{2i+1}{2} \Rightarrow i = \frac{2x-1}{2}$

Put this i in any tangent, we get $y = x^2 + bx + c - \frac{1}{4}$

49. Equation of tangent is $y = 4x \pm \sqrt{16a^2 + b^2}$ this normal passes through (–2, 0)
$\Rightarrow 0 = -8 \pm \sqrt{16a^2 + b^2}$; $264 = 16a^2 + b^2$
Giving AM ≥ GM
$\frac{16a^2 + b^2}{2} \geq \sqrt{16a^2b^2}$
$\frac{64}{2} \geq \sqrt{16a^2b^2}$
$4ab \leq 32$
$ab \leq 8$

50. Equation of tangent at A is $4x - 3y - 8 = 0$
Let $y = m(x - 2)$ a line thro (2, 0) and $m = -7$ or $\frac{1}{7}$
We have $\frac{x-2}{-\frac{1}{\sqrt{50}}} = \frac{y}{\frac{7}{\sqrt{50}}} = 5\sqrt{2}$
$\Rightarrow$ Centre of circle are (1, 7), (3, –7), (9, 1) and (–5, –1)

51. Normal at $P(x_1y_1)$
$y - y_1 = \frac{-1}{m}(x - k_1)$
Where $y = 0$, $x = x_1 + my_1$
We have $|x_1 + my_1| = 2|x_1|$

$\Rightarrow x_1 + my_1 = \pm 2x_1$

Case-I: $x_1 + my_1 = 2x_1 \Rightarrow m = \dfrac{x_1}{y_1}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{x}{y} \quad \left(\because m = \left(\dfrac{dy}{dx}\right)_{at\,(x_1y_1)}\right)$

$\Rightarrow x\,dx - y\,dy = 0;\ \dfrac{x^2}{2} - \dfrac{y^2}{2} = C - a$ hyperbola

Case-II: $x_1 + my_1 = -2x_1;\ m = \dfrac{-3x_1}{y_1}$

$y\dfrac{dy}{dx} + 3x = 0$

$\Rightarrow \dfrac{y^2}{2} + \dfrac{3x^2}{2} = C$ (Ellipse)

52. Given equation can be written as $\dfrac{4x}{x^4 + x^2 - 4} - \dfrac{4}{x} = 0$

$\Rightarrow x = \pm\sqrt{2}$

53. Equation can be written as $\cos 2x(\cos 4x + \cos 2x) = 0$
$\Rightarrow 2\cos x \cos 2x \cos 3x = 0$

54. Equation of normal to hyperbola $xy = c^2$ at $\left(ct, \dfrac{c}{t}\right)$ which passes through (h, k) is

$ct^4 - ht^3 + kt - c = 0$
Roots of this equation is t_1, t_2, t_3 and t_4
Let equation of circle be $x^2 + y^2 - 2gx - 2fy + p = 0$
If $\left(ct, \dfrac{c}{t}\right)$ lies on this $c^2t^4 - 2gct^3 + pt^2 - 2fct + c^2 = 0$

Its roots are $t_1\ t_2\ t_3$ and t_4'

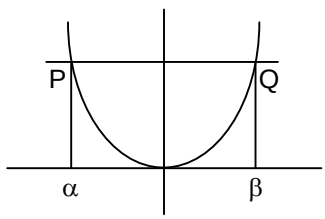
We get $t_4 = -t_4'$ also $\dfrac{c}{t_4} - \dfrac{c}{t_4'} = k - 2f$

So locus of centre is $4c^2 = (h - 2g)(k - 2f)$

Centre lies on hyperbola $\left(x - \dfrac{h}{2}\right)\left(y - \dfrac{k}{2}\right) = c^2$

$c\left(\dfrac{h}{2}, \dfrac{k}{2}\right)$ and $c = \sqrt{2}$

55.-56. Any two point on $y = x^2$ is $P(\alpha, \alpha^2); Q(\beta, \beta^2)$



Equation of PQ, $y - \alpha^2 = (\alpha + \beta)(x - \alpha)$
$y = (\alpha + \beta)x - \alpha\beta$

Required area $\int_{\alpha}^{\beta}\left((\alpha + \beta)x - \alpha\beta - x^2\right)dx$

$\Rightarrow \beta - \alpha = 2$
Pair of tangents from origin are $y = 2x$ and $y = -2x$

Area $\int_0^1\left(\left(x^2 + 1\right) - 2x\right)dx = \dfrac{2}{3}$

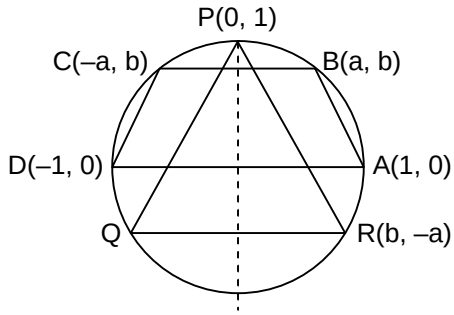
SECTION – C

57. As $\ell_1^2 + \ell_2^2 = (SS_1)^2$ (S, S_1 are focii of ellipse)

58. Suppose normals at α, β and γ are concurrent at (h, k) and let S be the foot of the fourth normals from (h, k), then we have, $\sum \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = 0$ and $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}\tan\frac{\delta}{2} = -1$

Eliminating $\tan\frac{\delta}{2}$ from above, we will get $\sin(\alpha + \beta) + \sin(\beta + \gamma) + \sin(\gamma + \alpha) = 0$

59. Let circle is $x^2 + y^2 = 1$



Equation of AB $y = \frac{b}{a-1}x + 1$ and $a^2 + b^2 = 1$

$\Rightarrow b^2 = (1 - a)(1 + a)$ or $\frac{b}{1-a} = \frac{1+a}{b}$

So equation of line $y = \frac{-1(1+a)}{b}x + 1$

So area of $\Delta = b + ab$
So ratio is 1

60. Apply AM $\geq$ GM

$$\frac{(\sin x)^{2y}}{(\cos x)^{\frac{y^2}{2}}} + \frac{(\cos x)^{2y}}{(\sin x)^{\frac{y^2}{2}}} = 2(\sin x \cdot \cos x)^{y - \frac{y^2}{4}}$$

It follows that $\sin 2x \geq 2(\sin x \cdot \cos x)^{y - \frac{y^2}{4}}$

$\because \sin x \cdot \cos x < 1 \Rightarrow 1 \leq y - \frac{y^2}{4}$ or $\left(1 - \frac{y}{2}\right)^2 \leq 0$

$\Rightarrow y = 2$ and $\sin x = \cos x$

So there is a unique solution $x - \frac{\pi}{4}$, $y = 2$

61. Let $P = \cos x \cos 2x \cdot \cos 3x$ ..... $\cos 999x$ and $Q = \sin x \sin 2x \sin 3x$ ..... $\sin 999x$
$PQ \cdot 2^{999} = \sin 2x \sin 4x \sin 6x$ ..... $\sin 1998x$
$= (\sin 2x \cdot \sin 4x$ ..... $\sin 998x)(-\sin(2\pi - 1000x)(-\sin(2\pi - 1002x)$ ..... $[-\sin(2\pi - 1998x)]$
$= \sin 2x \cdot \sin 4x$ ..... $\sin 998x \cdot \sin 999x \cdot \sin 997x$ ..... $\sin x = Q$

$P = \frac{1}{2^{999}}$ $(Q \neq 0)$

62. Equation of normal at $(t^2, 2t)$ is $y = -xt + 2t + t^3$
Let any point on $y = 2$ is (h, 2)
If normal passes through (h, 2)
$2 + ht - 2t - t^3 = 0$
$t^3 - t(h - 2) - 2 = 0$
t_1, t_2 and t_3 are points of this equation vertices lies on $xy = -2$

63. $$\frac{\sin(A + B - C) \cdot \cos(A - B + C)}{\sin(A - B + C)\cos(A + B - C)} = \frac{\sin C \cos B}{\sin B \cos C}$$

$$\Rightarrow \frac{\sin 2A + \sin(2B - 2C)}{\sin 2A - \sin(2B - 2C)} = \frac{\sin C \cos B}{\cos B \sin C}$$

Use C and D rule will get $\sin 2A + \sin 2B + \sin 2C = 0$

64. Let $P(x_1, y_1)$ equation of chord of contact $yy_1 = 8(x + x_1)$

Equation of line perpendicular to this and passes through P is $y - y_1 = \frac{-y_1}{8}(x - x_1)$

This touches $x^2 = 12y$

$\Rightarrow$ Locus is $8x + 3y + 64 = 0$

65. Let point $\left(2t_1, t_1^2\right)$ and $\left(2t_2, t_2^2\right)$ be the extremities of focal chord

$\Rightarrow t_1t_2 = -1$

Equation of focal chord $x(t_1 + t_2) - 2y + 2 = 0$ and line $y = mx + \sqrt{m^2 - \frac{1}{4}}$ is tangent to hyperbola passes through (0, a)

$\Rightarrow m = \pm\sqrt{1+\frac{1}{4}} = t_1 + t_2$

Length of focal chord = 9

66. Given equation can be written as $x^2 - 3 = 3\left[\sin\left(x - \frac{\pi}{6}\right)\right]$

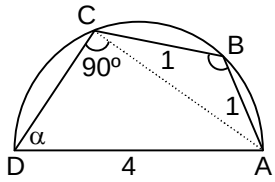
Hence, $x^2 - 3 = -3, 0, 3$

$\Rightarrow x = 0, \ x = \sqrt{3}$ is only solution

SECTION – D

67. Let $\angle ADC = \alpha$ and $AC = x$

By Cosine Rule



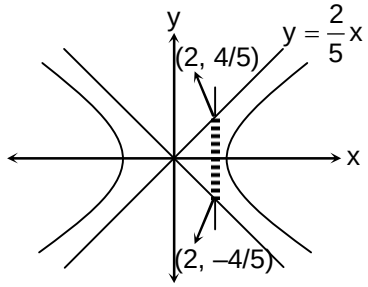
$x^2 = 1 + 1 - 2(1)(1)\cos(180^\circ - \alpha) = 2 + 2\cos\alpha = 2 + 2\cdot\frac{CD}{4}$

$= 2 + \frac{1}{2}\sqrt{16 - x^2} \Rightarrow x = \frac{\sqrt{15}}{2} \Rightarrow CD = \sqrt{16 - x^2} = \frac{7}{2}$

Perimeter $= 4 + 2 + \frac{7}{2} = \frac{19}{2} = 9.50$

68. Shown in the figure

$\alpha \in (-4/5, 4/5)$

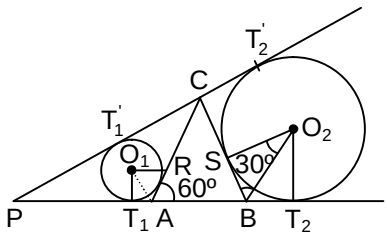


69. $\angle T_1O_1R = 60^\circ$ and $\angle AO_1R = 30^\circ$

Similarly $\angle BO_2S = 30^\circ$

$T_1T_2 = T_1A + AB + BT_1 = RA + RB + SB$

$= r_1 \tan 30^\circ + a + r_2 \tan 30^\circ = \frac{r_1 + r_2}{\sqrt{3}} + a$



and $T_1'T_2' = T_1'C + T_2'C = a - RA + a - BS = 2a - \frac{r_1 + r_2}{\sqrt{3}}$

Since common external tangents are equal $T_1T_2 = T_1'T_2'$

$\Rightarrow r_1 + r_2 = \frac{\sqrt{3}a}{2}$